Department of Art

University at Buffalo

Spring 2025

Interactive Art & Design

Instructor: Professor Marc Böhlen (mrbohlen@buffalo.edu)
ART 383 - Registration#: 21990, 3 credits
Location: CFA136
Times: Monday and Wednesday, 12:30 – 15:00
Office hours: Monday, 15:00

Course description:
This undergraduate course will introduce Art and Design students to interaction design, with a focus on conceptualizing and prototyping computationally mediated experiences. We will read texts from theorists and practitioners of interaction design and build toy systems that respond to topics and concepts introduced in the readings. Students will be asked to prepare short presentations on the assigned readings.

Interaction design is a wide field and we will use the online Encyclopedia of Human-Computer Interaction as reference (see below). From the many subfields of interaction we will focus on experience design, visualization, mobile and affective computing, context aware computing. We will also explore the use of AI assistants to support project development and coding. Students will learn the basics of Javascript while experimenting with generative AI models from ChatGPT and Anthropic and engage with collaborative design tools to support their research.

The semester project will comprise of two components: a simple game, designed for mobile phones and a prototype interactive website. Collaborations are encouraged. Details TBA.

Course materials:
All reading materials and assignments, including exam guidelines and preparations, will be available on UBlearns.
Encyclopedia of Human-Computer Interaction: https://www.interaction-design.org/literature

Coding languages:
Javascript and P5: https://editor.p5js.org/

Prototyping environments:
Figma: https://www.figma.com/

AI assistants:
ChatGPT: https://openai.com/chatgpt/
Anthropic: https://console.anthropic.com/dashboard
https://console.anthropic.com/workbench/

Prerequisites:
Curiosity, determination, web literacy, basic coding experience and chatbot exposure.

Deliverables:
a) 3 exercises
b) 1 group project
c) 1 semester project
d) 1 topic presentation / discussion lead

Details on the exercises and assignments will be shared in detailed documents available via UBlearns.

Grading:
a - 60%, b - 15%, c - 20% , d - 5% (total 100)

Grading scale:
100–90: A; 89–80: B; 79–70: C; 69–60: D (less than 60: F)
Class presence:
Students can miss at most 2 classes during the entire semester without penalty.
Three or more missed classes will result in a reduction of the course grade.

General learning objectives:
1) General understanding of interaction design concepts
2) Ability to apply interaction principles in design strategies

Assessment:
1) This knowledge will be tested in assignments # a, b
2) This ability will be tested in # b, c

Preliminary course schedule (subject to changes)

Monday classes will focus on readings and presentations. Wednesday classes will be dedicated to design activities.

Week	Topics	Exercises	Student co-presenters	Assignments	comments
W1	Course introduction	Intro ChatGPT - prompt design			
W2	History of interaction design	Introduction to Figma			
W3	Experience design	Introduction to Javascript			
W4	Data visualization	P5 data visualization		A1	
W5	Data visualization, 2	Big data visualization			
W6	Affective computing	Interaction concepts for intimate places			
W7	Mobile computing	JavaScript game on a mobile phone		A2	
W8	FALL BREAK				
W9	Interface design	Collaborations with Figma			
W10	Interface design, 2	Collaborations with Figma			
W11	Aesthetic computing	JavaScript game on a mobile phone		A3	
W12	Wearable computing	The human body as a source of data			
W13	Tactile interaction	Touch versus vision - data visualization			
W14	Sociology	Understanding people			
W15	Human-robot interaction	Robot companions		A4	
W16	Final project design				
				A5	

Learning Outcomes and Assessment

This course is designed for ambitious students. It requires above average time commitment and determination. Expect to invest one hour per day , 6 days a week to get an A grade.

The main learning outcomes each student should achieve are:

1) Interaction design literacy - Acquire a detailed overview of interaction design concepts.
2) Interaction design experience – Ability to intelligently work with interaction design concepts.

The table below describes the learning outcomes and assessment approaches on a topic by topic level.

Topic	Learning Outcomes	Assessment
1. Interaction design literacy	Understanding of fundamental ideas in interaction design.	Assignments 1, 2, 3
2. Interaction design experience	Application of interaction design to creative or critical exploration.	Group and Final project

UB Portfolio

If you are completing this course as part of your UB Curriculum requirements, please select an 'artifact' from this course that is representative of your learning and save it in a safe location with a clear title. Your final UB Curriculum requirement, UBC 399: UB Curriculum Capstone, will require you to submit these 'artifacts' as you process and reflect on your achievement and growth through the UB Curriculum. Artifacts include homework assignments, exams, research papers, projects, lab reports, presentations, and other coursework. For more information, see the UB Curriculum Capstone website: https://www.buffalo.edu/ubcurriculum/capstone.html.

Relevant UB Policies

- Academic Integrity
  Students are required to comply with the University's policy on Academic Integrity, found on the university website: https://academicintegrity.buffalo.edu/policies.php. The following actions, among others listed in the policy, constitute academic dishonesty: Work submitted to other courses; Plagiarism; Receiving major assistance completing an assignment without acknowledging that assistance; Falsification of academic materials; Misrepresentation of documents; Selling or purchasing academic assignments. The sources of all quoted or appropriated material must be credited.

- Accessibility
  UB respects and welcomes students of all backgrounds and abilities. In the event you encounter any barrier(s) to full participation in this course due to the impact of disability, please contact the Accessibility Resources Office. The access coordinators in the Office of Accessibility Resources can meet with you to discuss the barriers you are experiencing and explain the eligibility process for establishing academic accommodations. You can reach the Office of Accessibility Resources through www.buffalo.edu/studentlife/accessibility
  (716) 645-2608; 60 Capen Hall.

- Wellness
  Sexual Violence: UB is committed to providing a safe learning environment free of all forms of discrimination and sexual harassment, including sexual assault, domestic and dating violence and stalking. If you have experienced gender-based violence (intimate partner violence, attempted or completed sexual assault, harassment, coercion, stalking, etc.), UB has resources to help. This includes academic accommodations, health and counseling services, housing accommodations, helping with legal protective orders, and assistance with reporting the incident to police or other UB officials if you so choose. Please contact UB's Title IX Coordinator at 716-645-2266 for more information. For confidential assistance, you may also contact a Crisis Services Campus Advocate at 716-796-4399.

- Mental Well-Being

  As a student you may experience a range of challenges that can cause barriers to learning or reduce your ability to participate in daily activities. These might include strained relationships, anxiety, high levels of stress, alcohol/drug problems, feeling down, health concerns, or unwanted sexual experiences. Counseling, Health Services, and Health Promotion are here to help with these or other issues you may experience. You learn can more about these programs and services by contacting:

  Counseling Services:
  120 Richmond Quad (North Campus), 716-645-2720
  202 Michael Hall (South Campus), phone: 716-829-5800
  Health Services: Michael Hall (South Campus), 716- 829-3316
  Health Promotion: 114 Student Union (North Campus), 716- 645-2837

- Emergencies

  In case of emergency, call University Police, 645-2222.
  A first aid kit is located in CFA 142. An AED is located in the hallway opposite CFA 136 and elsewhere throughout most buildings on campus. In the event of a medical emergency, confirm that the environment is clear of hazards, begin first aid/rescue if qualified, have someone retrieve the AED and phone 645-2222. AEDs (automated external defibrillators) are critical life saving devices used in cases of cardiac arrest, and must be used by trained personnel only. Several CFA staff in the EP/GD area have been trained in the operation of the AED, including Daniel Calleri (CFA 123), Dom Licata (CFA 139) and Vince Harzewski (CFA 103).